MODULE 1: SENSORS

Introduction

Developments in Science and Technology during last century has resulted accelerated growth in emissions of anthropogenic pollutants in to environment contributing to increasing incidences of chronic diseases. **Toxic gases, metals, pesticides, and other dangerous chemicals have spread throughout our environment.** Therefore, monitoring of composition of air, water and soil has become an area of intense interest. In health sector, for diagnosis and health monitoring, it is required to determine clinically important chemicals present in body fluids. For various industrial and agricultural applications qualitative detection and quantitative analysis of chemical species is the major requirement. These demands have led to the development of many important analytical approaches. These techniques are not suitable for routine and online analysis. Therefore there is a growing demand to develop new instruments, techniques, and procedures so as to make analysis convenient, fast, reliable, and selective

To meet these demands, chemical sensors of various types have been intensively developed; these sensors have number of advantages over other chemical method of analysis like **low cost, less time analysis, portability of the device, accuracy of the results** etc.

Definition: A sensor is a device that can detect and measure specific chemical or physical parameters in a given environment.

Chemical sensor is a device that is capable of giving real time **analytical information** about a test sample. It interact with specific chemical or biological

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analyte**,** detects it and produces the signal proportional to its quality/quantity. Analyte is the target/test species which is being detected and measured using a sensor

Example: **mercury thermometer,** here the quantity that is being measured is heat or temperature. The measured temperature is converted to a readable value on the calibrated glass tube, based on the expansion and contraction of liquid mercury

Basic components of sensors: 1. Receptor, 2. Transducer, 3. electrical signal and display

1. **Receptor** → Receptor is chemical or biological recognition element, which is capable of interacting with analyte specifically and selectively. It produces a signal corresponding to interaction in the form of change in potential, conductivity, current, mass, heat, pH, color etc. If the receptor is biological origin (i.e., DNA, antibodies and enzymes) the device is referred to as **biosensors**. If the receptor is of chemical origin, the device is referred to as a **chemical sensor**.

2. **Transducer**→ are the devices that convert energy in one form into another form. Generally energy in the form of signal (example: thermometer, microphone, speaker etc.). In most cases sensing element and transducers are packed together.

3. **Electrical signal and display** → the electronic system analyses the signal given by transducer, helps in signal amplification and converts the signal from analog to digital form. These amplified signals are then displayed, signal can be displayed in various form such as numeric value, graph, image etc.

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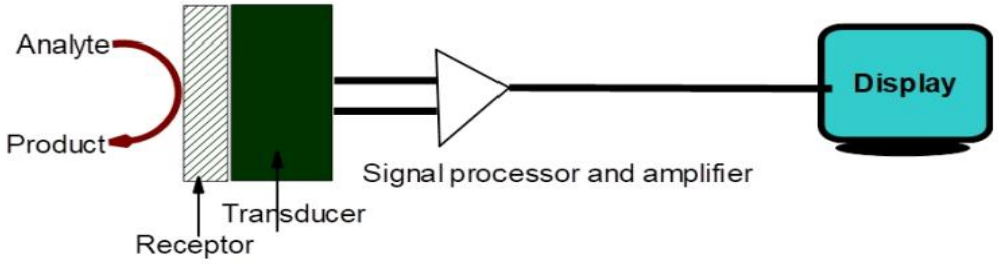

Fig. A schematic diagram illustrating the major components of a standard sensor.

Interaction between the receptor and analyte can be monitored by several methods. Different parameters are measured in each method like change in potential, conductivity, current, mass, heat, pH, colour etc. For measurement of each parameter, a different type of transducer is used. There are several types of sensors available depending on type of transducer is used like Electrochemical sensors, Conductometric sensors, Thermometric sensors, and Optical sensors etc.

Actuators: Actuators are the devices that work opposite to sensors. A sensor converts a physical event into an electrical signal, whereas an actuator converts electrical signal into a physical event. When sensors are used at the input of a system, actuators are used to perform output functions in a system while controlling an external device.

1. Electrochemical sensors

Definition: Electrochemical are the devices that will convert chemical information into electrical signal suitable for a computer or a monitor. Electrochemical sensors use an electrochemical reaction to detect and measure the concentration of the analyte in a sample.

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Construction: In Electrochemical sensors the working/sensing electrode is used as transducer element.

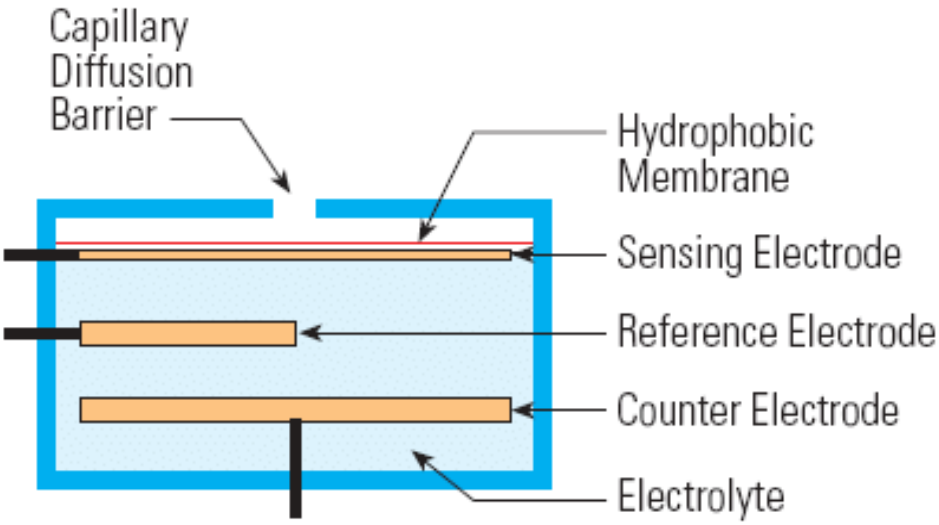

Fig. Block diagram of electrochemical sensor device

- Electrochemical sensor consists of a transducer element covered by a **recognition element**. The recognition element interacts with target analyte and signal is generated.
- The sensing electrode has a **chemically modified surface** (depending on the application, chemicals are used to modify the surface of the electrode), ensure the selectivity and facilitating the reduction / oxidation of analyte.
- The device uses an **electrolyte** that transports the charge within the sensors and effectively contacts all electrodes. Also, it dissolves the reactant and product for efficient transport and remains chemically and physically stable under all sensor operating conditions.
- A closed circuit is always required for electrochemical measurements; current must flow in order to be able to carry out a measurement. Therefore, a counter electrode closes the circuit and balances the electrochemical redox reactions.

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Working:

The following steps are involved in the working of electrochemical sensors

1. Diffusion (*Diffusion is the net movement of anything (for example, atoms, ions, molecules, energy) generally from a region of higher concentration to a region of lower concentration*) of the analyte occurs towards the interface between the electrode and the electrolyte (in the liquid phase)
2. Adsorption (*Adsorption is the process by which ions, atoms or molecules adhere to the surface of a solid material.*) occurs on to the electrode surface
3. Electrochemical reactions (oxidation/reduction) takes place with electron transfer from anode to cathode.
4. Desorption (*Desorption is the physical process where adsorbed atoms or molecules are released from a surface into electrolyte or gaseous phase*) of the products
5. Diffusion of the products away from the reaction zone to electrolyte or gas phase

Applications

1. Electrochemical sensors used for the detection of blood glucose
2. Electrochemical sensors are used for pH measurements
3. Used to detect pesticides
4. Used in the detection of hydrocarbon pollutants
5. Biomedical bacterial and viral diagnosis
6. Soil parameter analysis in agriculture

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2. Electrochemical Sensor for the measurement of dissolved oxygen (DO); by Galvanic oxygen sensor

Definition and principle: Galvanic oxygen sensors, also known as **amperometric** or **Clark-type sensors**, measures dissolved oxygen concentration in water based on electrical current produced. **Galvanic Sensor is a Electrochemical sensor, which is used to measure DO.**

Construction:

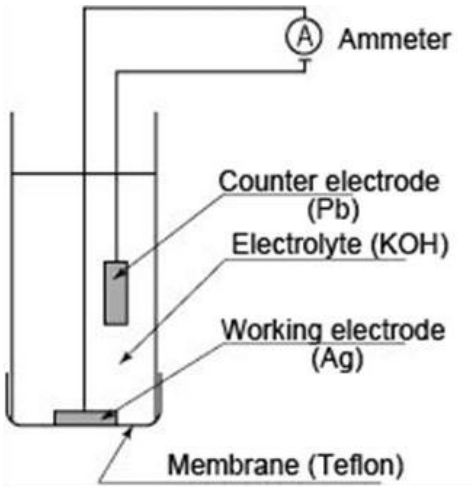

Fig. Galvanic Sensor device for measurement of DO.

The following components are involved in the device:

- The cathode is a working electrode made of Ag, Pt, or Au metals
- The anode is made of the active metals such as Zn, Pb or any other active metals
- The electrolyte is used as KOH, NaOH or any other inert electrolyte (acetate)
- The membrane in the bottom surface of the working electrode consist of a **Teflon membrane** is a gas permeable membrane that is selective for oxygen

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Working:

1. The anode and cathode are two dissimilar metals immersed in a analyte sample containing dissolved oxygen, the difference in potential between the anode and the cathode should be at least 0.5V, in order to reduce dissolved oxygen (O_2^{2-}) to hydroxide ion (OH^- ion).
2. Oxygen molecule diffuses across the oxygen-permeable membrane (Teflon) and the rate of diffusion is proportional to the amount of oxygen in the analyte.
3. The molecular Oxygen (O_2^{2-}) reduces to OH^- at cathode (reaction mentioned below). This reaction produces an electrical current that is directly proportional to the oxygen concentration. This current is carried by the ions in the electrolyte and runs from the **cathode to the anode.**
4. Reactions:

$$\text{Anode(Pb):}\quad \text{Oxidation}\quad 2Pb \longrightarrow 2Pb^{2+} + 4e^-$$

$$\text{Cathode(Ag):}\quad \text{Reduction}\quad O_2 + 4e^- + 2H_2O \longrightarrow 4OH^-$$

$$\text{Overall:}\quad 2Pb + O_2 + 2H_2O \longrightarrow 2Pb(OH)_2$$

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Disposable screen printed sensors

Disposable Screen-printed electrodes are electrochemical measurement devices designed for single point usage that are manufactured by printing **the conductive ink** on PVC substrate or plastic or ceramic substrates, allowing quick in-situ analysis with high reproducibility, sensitivity and accuracy.

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Advantages of disposable screen printed sensors:

1. Disposable sensors are biodegradable and sustainable

2. They have a short duration of analysis and fast response times.

3. It provide digitized chemical and biological information.

4. Prevents the contamination of samples.

3. Detection of Ascorbic acid using disposable electrochemical sensor

L-Ascorbic acid (AA), commonly known as vitamin C, is an important water-soluble vitamin derived from green vegetables, fruits, and other dietary supplements. AA is responsible to maintain the bone health, teeth health, skin health etc. **At the same time** Abnormal AA levels in bodily fluids have been reported to cause cancer, cardiovascular diseases, and Alzheimer's and Parkinson's diseases. Extended use of AA could cause urinary oxalate calculus, increase infertility in a woman, and affect embryo development.

Construction:

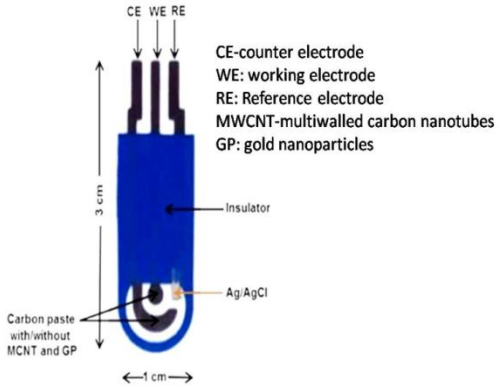

Fig. screen printed electrode diagram

The following components are involved in the device:

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- The working electrode (WE) made of carbon paste (graphite sheet) with MWCNTs & gold nanoparticles.
- The counter electrode (CE) is made of carbon paste (graphite sheet) with MWCNTs & gold nanoparticles.
- Reference electrode (RE) is made of Ag/AgCl

Working

1. The electrochemical sensor can catalyze the **two-electron electro catalytic oxidation and hydrogen dissociation of AA to L-dehydroascorbic acid in the presence of oxygen in solution.**

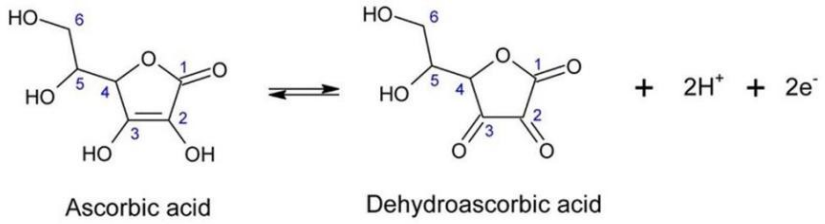

Fig: 1.18 Oxidation of Ascorbic acid

2. The carboxyl MWCNTs act as electron transfer mediators promoting the electron transfer between AA molecules and the matrix interface Increases the electro catalytic oxidation of AA during electrochemical detection and transduce their presence into measurable signals.

3. The electric current produced is proportional to the concentration of the dehydroascorbic acid.

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